

Abhishek HS

LLM Systems · Efficient Fine-tuning · Agentic AI · Medical AI · Graph Neural Networks

+91-8660210830 | abhiaklapura@gmail.com | Bangalore, India | GitHub | Google Scholar

EDUCATION

Bachelor of Technology — Computer Science Engineering (AI & ML)

2022 – 2026

PES University, Bangalore · CGPA: 7.6 / 10.0 · Relevant Coursework: Deep Learning, Computer Vision, NLP, Graph Neural Networks, Algorithms

PUBLICATIONS

[ACM BCB 2026]

“Transformer-Guided Graph Attention for Direct Cardiac Mesh Reconstruction: A Structural Digital Twin Framework.”

Accepted

ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM BCB), 2026.

First Author

[ICML Workshop 2026]

“ReTreVal: Reasoning Tree with Validation for Mathematical Problem Solving.”

Accepted

ICML Workshop on LLM4Plan, 2026.

First Author

Both papers accepted; proceedings forthcoming.

RESEARCH EXPERIENCE

Research Intern — AI/ML

May 2024 – Sep 2024

CAVE LABS, PES University · Bangalore

- Engineered an end-to-end pipeline — raw CT/MRI → 3D Swin Transformer segmentation → Graph Attention Network (GAT) mesh deformation → structured cardiac digital twin — achieving 0.84+ Dice score across 6 cardiac structures; results accepted at ACM BCB 2026 (First Author).
- Replaced traditional marching cubes post-processing with direct surface prediction, reducing mesh reconstruction time by 60% while maintaining anatomical consistency across structures including pulmonary artery (0.80 Dice).
- Led a 3-member research team; managed full research lifecycle from ideation to paper submission.

INDUSTRY EXPERIENCE

AI Engineer Intern

Oct 2025 – Present

QpiAI · On-Site, Bangalore

- Researched and applied Group Relative Policy Optimization (GRPO) for reinforcement learning on LLMs; investigated reward shaping strategies to align small models (2B–7B parameters) with task-specific objectives without human annotation overhead.
- Designed and trained specialized sub-agents using compact models (Qwen 2B, LLaMA 7B) within the AgentHive multi-agent framework, assigning targeted subtasks to small models to reduce inference cost while preserving end-to-end task performance.
- Implemented an NLP-to-SQL pipeline for the AgentHive Generalist Agent, enabling natural language querying of enterprise relational databases and closing the loop between structured data and autonomous decision-making.
- Researched and implemented advanced prompting strategies (CoT, ReAct, Reflexion, Self-Refine) to enhance LLM reasoning; authored internal knowledge base on LLM optimization and agentic design patterns.

KEY PROJECTS

FIT LLM — Fault-tolerant Inference and Training on Three-Tier Memory Hierarchy

PyTorch · CUDA · NVMe SSD · PCIe DMA · LoRA · AdamW · HBM

- Designed a **three-tier memory hierarchy** (NVMe SSD → CPU RAM → GPU VRAM/HBM) enabling full fine-tuning of **70B parameter LoRA models** with **peak GPU VRAM under 4 GB** — making large-scale LLM training feasible on commodity hardware.
- Engineered a **shard-wise forward and backward loop**: each transformer shard loaded on-demand via **PCIe DMA** (NVMe → GPU), forward-passed on GPU, checkpointed to CPU RAM, and gradient-accumulated back to NVMe — eliminating the need to hold full model weights in VRAM simultaneously.
- Implemented **async NVMe prefetching** (shard $i+1$ pre-loaded while shard i computes) and **overlapped PCIe/GPU/NVMe transfers**, saturating the full **3.5 GB/s NVMe** and **100+ GB/s DDR5** bandwidth hierarchy.
- Built **per-shard SHA-256 checksum** verification and activation checkpointing ($h_0 \dots h_N$ stored in CPU RAM) for fault tolerance; all LoRA adapter matrices (~320 MB for rank-16 on 70B) kept resident in CPU RAM across training steps.
- Integrated **AdamW optimizer state offloading** (FP32 master weights + momentum tensors) to NVMe, reducing total memory footprint from ~50 GB to ~12 GB (4-bit) with gradient accumulation across batches.

TECHNICAL SKILLS

Research Areas	LLM Systems, Efficient Fine-tuning, RL for LLMs (GRPO), LLM Reasoning, Medical AI, Graph Neural Networks
LLM & Transformers	LLaMA, Qwen 2.5, GPT, BERT — LoRA, QLoRA, RLHF, GRPO, fine-tuning, attention mechanisms, quantization
Systems & Hardware	CUDA, PCIe DMA, NVMe I/O, HBM, gradient checkpointing, optimizer offloading, 4-bit quantization
Agentic Frameworks	LangChain, LangGraph, LlamaIndex — memory, planning, tool use, Tree-of-Thoughts, ReAct, Reflexion
ML & DL Frameworks	PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, OpenCV
Medical AI	3D Swin Transformer, Graph Attention Networks, CT/MRI segmentation, mesh deformation
Programming	Python (primary), C, CUDA C, SQL

HONORS & ACTIVITIES

Mentored 20+ teams on AI/ML projects; evaluated 50+ technical solutions at PES University events; improved average project quality scores by 30%	2023 – 2025
Logistics Head, Appex PESU — managed Horcrux 24-hr ideathon (100+ participants, \$10K+ sponsorships)	Apr 2024 – Apr 2025